

1. **Geometric units:** In the geometric unit system, $G = c = 1$. Under this convention, the physical quantities mass, length and time can be expressed in the same units.

- (a) Calculate the mass of the Sun ($1M_{\odot}$) in meters and in seconds.
- (b) Calculate the Schwarzschild Radius R_S for the Sun.
- (c) The supermassive black hole at the center of our galaxy, Sagittarius A*, has a mass of approximately $4.3 \times 10^6 M_{\odot}$. Determine the Schwarzschild radius R_S of Sagittarius A*.

2. (a) Starting from the contravariant form of the geodesic equation which we derived in class:

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma^{\mu}_{\alpha\sigma} \frac{dx^{\alpha}}{d\tau} \frac{dx^{\sigma}}{d\tau} = 0$$

obtain the following covariant form of the equation of motion:

$$m \frac{dp_{\beta}}{d\tau} + \Gamma^{\gamma}_{\beta\alpha} p^{\alpha} p_{\gamma} = 0$$

- (b) For the Schwarzschild metric, using the covariant geodesic equation derived in part (a), find the equations of motion for the components $\beta = 0$ and $\beta = 3$.
3. In class we saw that for the Schwarzschild metric, the radial motion of a particle in the equatorial plane ($\theta = \pi/2$) is described by the following effective potential:

$$V_{\text{eff}}(r) = -\frac{M}{r} + \frac{\ell^2}{2r^2} - \frac{M\ell^2}{r^3} \quad (1)$$

- (a) Show that in the Newtonian limit, the Schwarzschild effective potential reduces to the standard Keplerian form.
- (b) Find the values of r for which $V_{\text{eff}}(r) = 0$.
- (c) Obtain the condition for the circular orbits.
- (d) Show that the $r = 6M$ corresponds to the lowest stable circular orbit. Convince yourself that it is indeed the smallest orbit and hence referred as Innermost stable circular orbit (ISCO)

4. We showed in the class that the motion of the test particle in Schwarzschild space-time governed by mass M can be given by

$$\frac{\tilde{E}^2}{c^2} = \left(\frac{dr}{d\tau} \right)^2 + V^2$$

where

$$V^2(r) = c^2 \left(1 - \frac{2GM}{rc^2} \right) \left(1 + \frac{\ell^2}{r^2 c^2} \right) \quad (2)$$

and $\tilde{E} = E/m$ and ℓ is the angular momentum per unit test mass m .

- Express $V^2(r)$ in geometric units. Obtain the conditions for optimum values. Please understand that it crucially depends on the ℓ/M ratio.
 - Using your favourite programming language/software, plot $V^2(r)$ as a function of r . Study various cases: $\ell > 4M$, $\ell = 4M$, $\ell = 2\sqrt{3}M$, and $\ell < 2\sqrt{3}M$,
 - Obtain conditions of possible trajectories for all the above cases.
5. • Calculate the non-zero components of the $R_{\mu\nu\sigma\delta}$ in the Schwarzschild spacetime and show that the Ricci scalar is zero.
- **Kretschmann scalar:** Verify that the Kretschmann scalar $R_{\mu\nu\sigma\delta}R^{\mu\nu\sigma\delta}$ is finite at the event horizon.
6. If no light can escape the BH, how do we detect these systems? What was the first observational signature of these objects?
7. Consider the following metric that describes a weak gravitational field:

$$ds^2 = \left(1 + \frac{2\phi}{c^2} \right) dt^2 - dx^2 - dy^2 - dz^2 \quad (3)$$

where $\phi = \phi(x, y, z)/c^2 \ll 1$. Calculate Riemann curvature tensor, Ricci tensor and Ricci scalar up to the lowest non-trivial order in ϕ/c^2 .